Exploratory Data Analysis

Assessment 1 · Group030

FIT5196 Data Wrangling

King Man Chan · Yinglin Fang · Sizhe Hong

Xinhang Ren · Guohou Zhang

1. Data and analytical approach

1.1 Start from the submitted relational data

The analysis begins by loading the six standardised CSV files produced in the solution notebook. The tables contain 5,000 orders, 15,723 order-item lines, 500 customers, 5,000 deliveries, 1,000 products and 7,000 reviews. Identifiers remain strings; only the numeric, date and Boolean fields used below are converted after loading.

Table	Rows	Columns	Role in the EDA
orders	5,000	23	Order value, discount and time
order_items	15,723	6	Product-level sales contribution
customers	500	20	Prior customer activity
deliveries	5,000	20	OTIF and fulfilment
products	1,000	21	Category and catalogue cost
product_reviews	7,000	21	Rating and review timing

1.2 Check the data before analysis

Before choosing the figures, we rechecked that the row count of each table matched its primary-key count. All six checks passed. For relational figures, the parent key was required to be unique and the left-hand row count had to remain unchanged. Figure 2 retains order grain for its group comparisons and aggregates to customer grain only for its customer scatter. Figure 8 retains review grain and separately reports the number of distinct reviewed orders. These checks protect the observation unit used in each chart.

Preparation decision	Implementation	Validation evidence
Structured extraction first	Parse JSON/XML before regex cleaning.	`VAL-SCHEMA-01–07`: six exact schemas pass.
Standardise before reconciliation	Compare converted fields by stable keys.	`VAL-DUP-01`, `VAL-OVERLAP-01`, `VAL-CONFLICT-01`: separate counts; zero conflicts.
Retain relational grains	Keep six entity tables and one-to-many links.	`VAL-PK-01–06`, `VAL-FK-01–08`: complete keys; zero orphans.
Apply published arithmetic and time order	Round line revenue before order totals; derive delivery timing.	`VAL-ARITH-00–03`, `VAL-TIME-03`: $0.00 maximum difference; zero timing conflicts.
Apply field-specific text rules	Lower-case three designated narratives; preserve delivery-note case; remove published wrappers and emoji.	`VAL-TEXT-02, 09–12`: 303 non-Latin reviews retained; case checks pass; wrapper and emoji residue are zero.

1.3 Move from table structure to analytical questions

We first examined order-value distribution and composition, then customer groups, product categories and time. We next tested review behaviour, multilingual text measurement, delivery operations and the delivery-review relationship. This sequence progresses from single-table description to checked relational analysis.

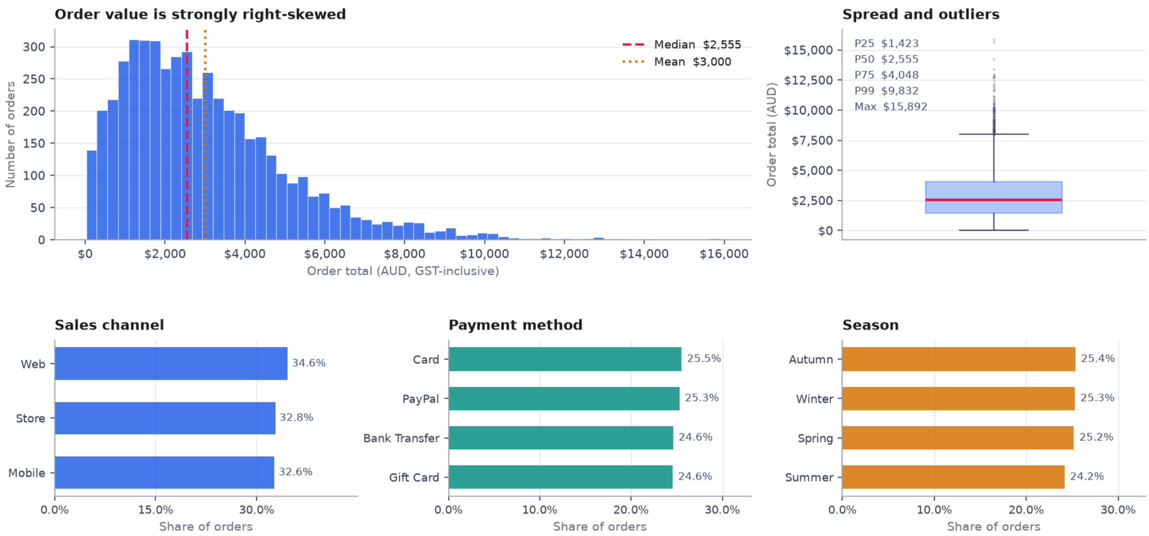
Continuous group means are reported with 95% confidence intervals. OTIF panels show their observed denominator and the report states the relevant uncertainty or alternative explanation. Results are stated at their observed grain; causal evaluation would require a separate research design.

1.4 Assessed EDA coverage

Figures 1–8 collectively cover univariate distribution/composition, bivariate comparison, multivariate/segmented analysis, temporal patterns, review/text behaviour and delivery/operational performance. Figures 2, 3, 7 and 8 are checked relational analyses.

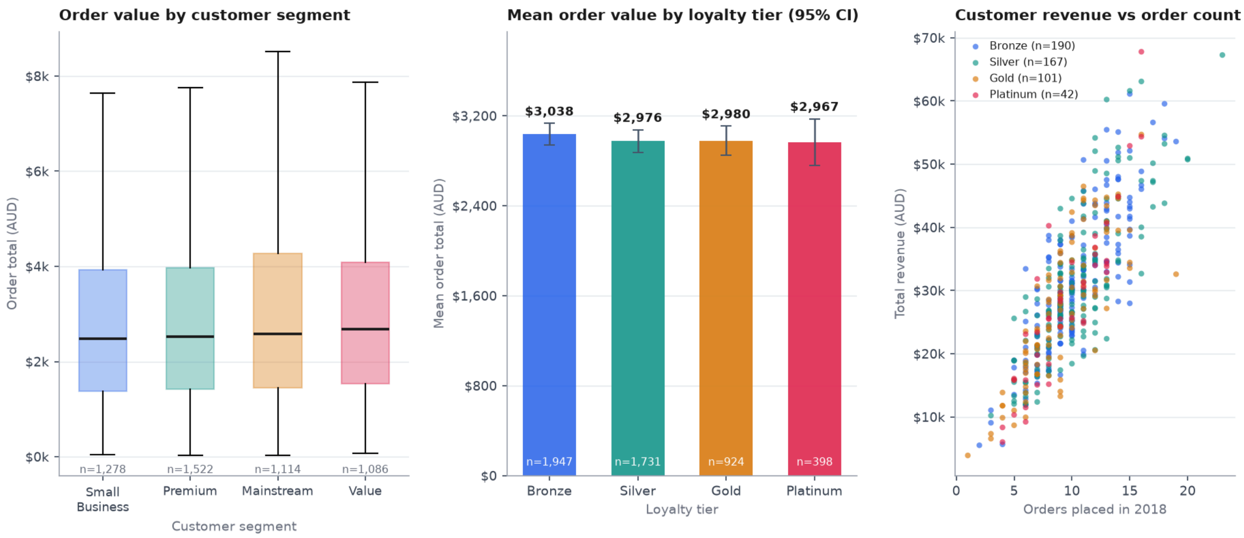
2. Assessed visualisations

2.1 Figure 1 — How are order values distributed, and how is order volume composed?



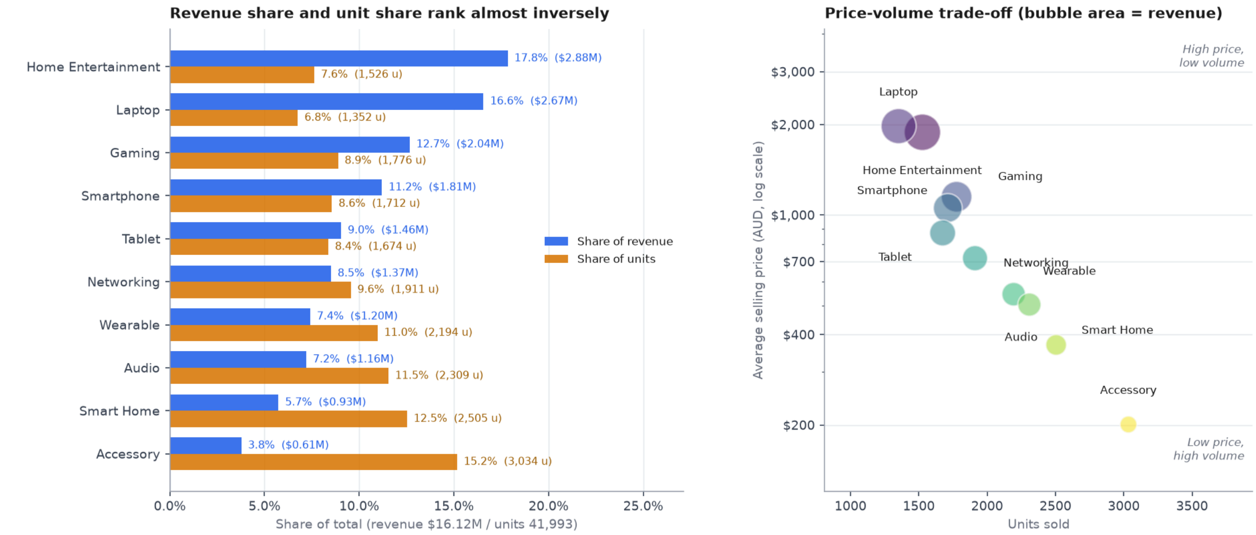
Observation unit and denominator. One canonical order; 5,000 orders. **Tables and join keys.** `orders`; no join. **Why this visualisation.** A histogram and boxplot retain the skew and outliers, while aligned percentage bars compare three compositions using a common order denominator. **Interpretation and explanation.** The median is more representative than the mean because a small high-value tail pulls the mean upward. Near-uniform channel, payment and season shares show that order counts are not concentrated in one route. **Material limitation.** Order totals combine baskets containing different numbers of items, and equal order shares do not imply equal revenue shares.

2.2 Figure 2 — Do customer segment and loyalty tier explain order value?



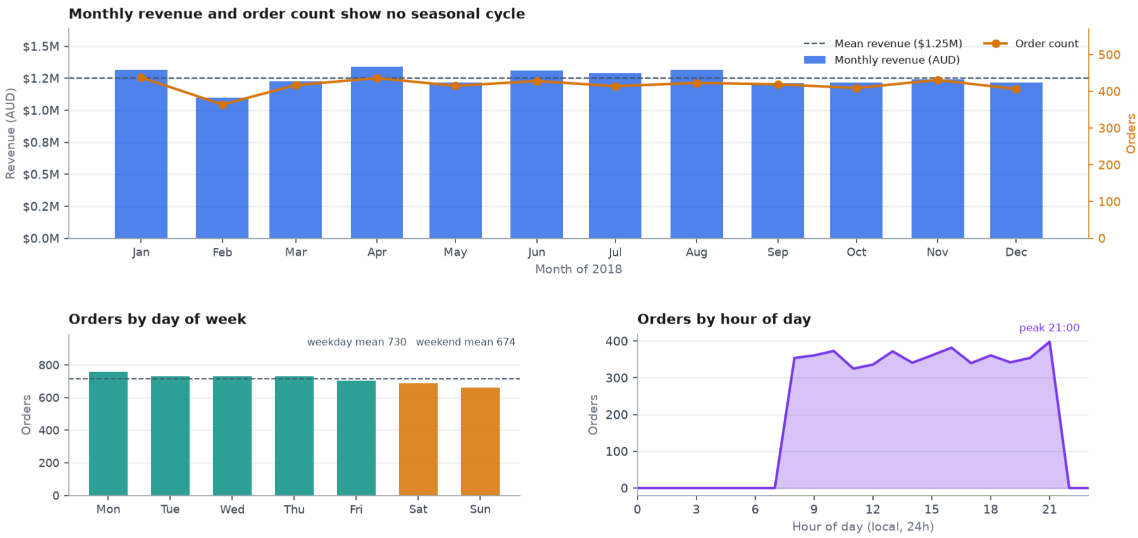
Observation unit and denominator. Left and centre: one order, 5,000 orders; right: one customer, 500 customers. **Tables and join keys.** `orders` joined to `customers` on `customer_id` (many-to-one); the order count must remain 5,000. **Why this visualisation.** Boxplots show the full within-segment spread, mean points/bars with 95% intervals show uncertainty by tier, and a customer-level scatter separates cumulative revenue from order frequency. **Interpretation and explanation.** Customer segment and loyalty tier do not meaningfully separate order value in this dataset. Customer revenue is instead closely aligned with the number of orders recorded during 2018. **Material limitation.** Customers contribute repeated orders, and loyalty tier is recorded at export rather than at the time of each order.

2.3 Figure 3 — Which categories drive revenue, and is that volume or price?



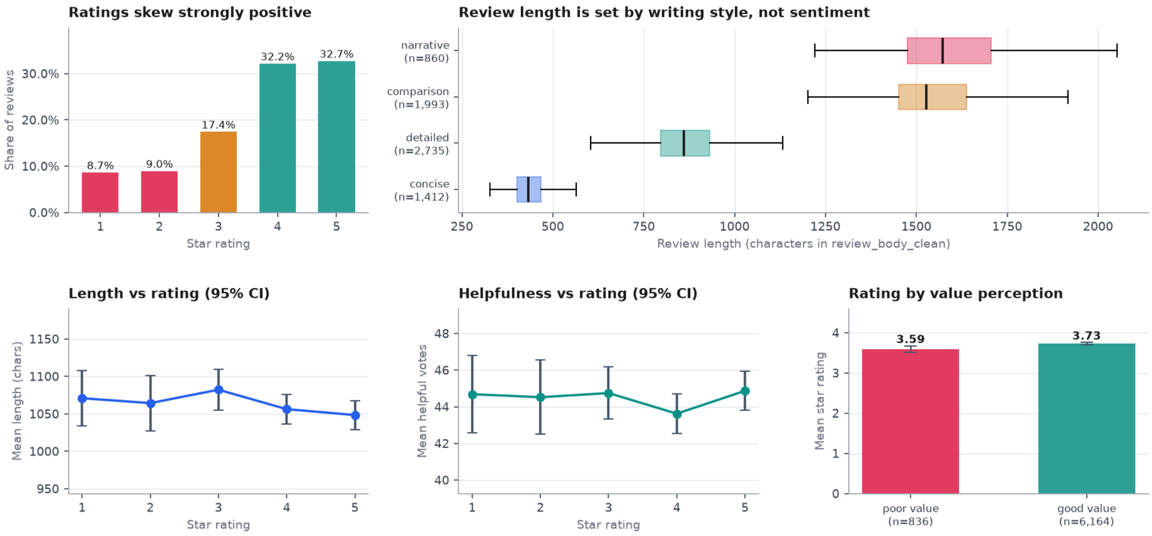
Observation unit and denominator. One order-item line; 15,723 lines, 41,993 units and $16.12 million line revenue. **Tables and join keys.** `order_items` joined to `products` on `product_id` (many-to-one); row count and revenue are checked after joining. **Why this visualisation.** Paired horizontal bars compare revenue and unit shares on the same scale. The price–volume bubble chart adds average selling price and revenue without summing order-level fields at item grain. **Interpretation and explanation.** High-priced categories generate large revenue shares from relatively few units, whereas accessories have high volume but low revenue share. Price point therefore explains much of the category ranking in this period. **Material limitation.** Revenue is not profit; cost, returns and promotions could change the commercial ranking.

2.4 Figure 4 — How does demand move through the year, the week and the day?



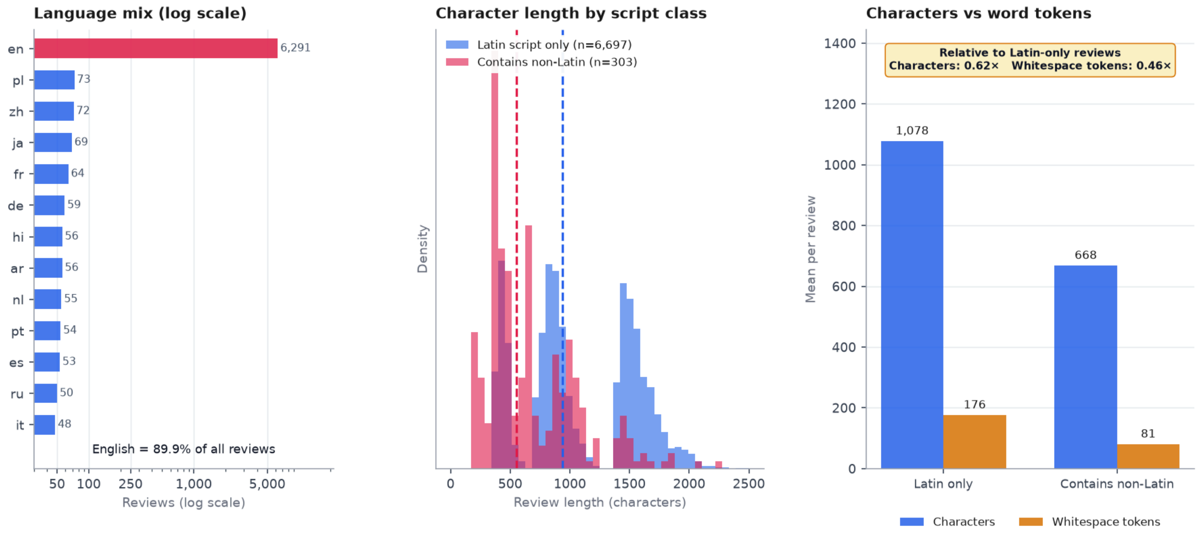
Observation unit and denominator. One order aggregated by month, weekday or hour; 5,000 orders from 1 January to 31 December 2018. **Tables and join keys.** `orders`; no join. **Why this visualisation.** Coordinated time panels distinguish annual, weekly and intraday structure. Bars make period totals readable and the hourly area-line plot shows concentration through the day. **Interpretation and explanation.** Month and weekday volumes vary little, while orders concentrate in trading hours and fall sharply overnight. Intraday timing is therefore more relevant to short-term staffing than a seasonal narrative. **Material limitation.** The data covers one calendar year, and the timestamps do not include a timezone.

2.5 Figure 5 — What drives review length, helpfulness and star rating?



Observation unit and denominator. One canonical product review; 7,000 reviews. **Tables and join keys.** `product_reviews`; no join. **Why this visualisation.** A rating composition chart, writing-style boxplots and mean comparisons with 95% intervals distinguish distribution, text length and labelled experience. **Interpretation and explanation.** Ratings are strongly positive. Review length differs sharply by supplied writing style but changes little across ratings, and helpful votes are similarly flat. The supplied value-experience label has only a modest rating difference. **Material limitation.** Writing style and value experience are source labels, not independently inferred text features, while helpful votes lack a views/exposure denominator.

2.6 Figure 6 — Does script class distort the review-length measures?

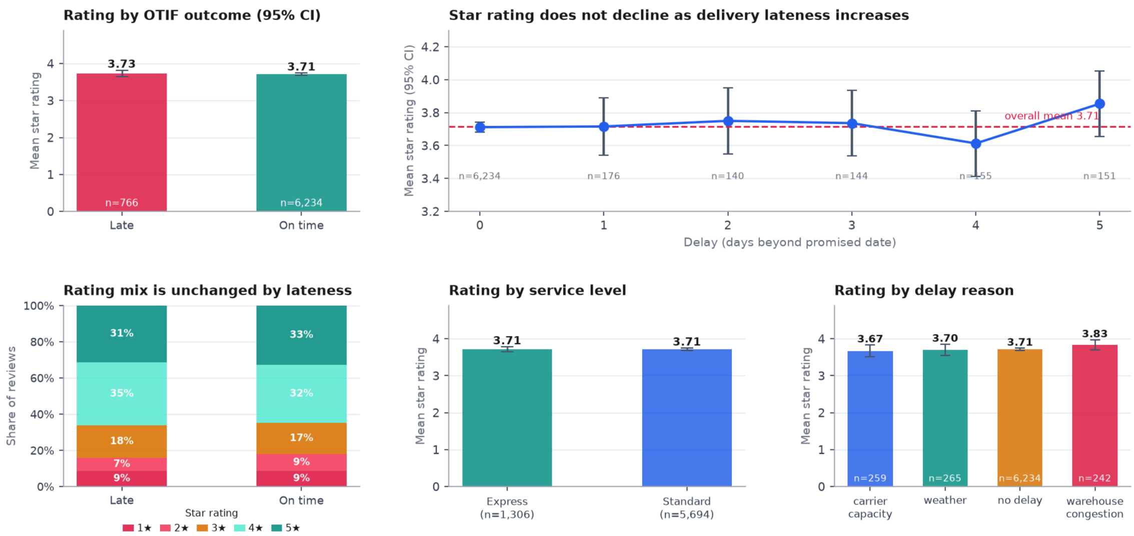


Observation unit and denominator. One review; 7,000 reviews, including 303 containing non-Latin script. **Tables and join keys.** `product_reviews`; no join. **Why this visualisation.** A log-scale language count handles the large English/non-English imbalance; overlaid distributions and paired means compare character and whitespace-token measures. **Interpretation and explanation.** Character count and whitespace-token count behave differently for non-Latin reviews. Character length provides the more consistent cross-script comparison, while language-aware tokenisation is needed for word-based analysis. **Material limitation.** The binary script flag groups several writing systems together, and the non-Latin group contains 303 reviews.

2.7 Figure 7 — Where does delivery performance actually vary?

Observation unit and denominator. One completed delivery; 5,000 deliveries, of which 536 (10.7%) are late. **Tables and join keys.** `deliveries` joined to `orders` on `order_id` (one-to-one); row count remains 5,000. **Why this visualisation.** Grouped bars compare carrier/service OTIF rates, while coordinated panels test delay reasons, warehouse distance, fulfilment by distance and expedited requests. **Interpretation and explanation.** OTIF varies by only a few percentage points across carriers and service levels; delay reasons are near-even and fulfilment averages remain close across the observed distance bands. There is no single dominant operational split in this export. **Material limitation.** The recorded distances cover a narrow metropolitan range, and carriers serve different mixes of orders.

2.8 Figure 8 — Does delivery performance shape customer ratings?



Observation unit and denominator. One review; 7,000 reviews covering 4,044 distinct orders. **Tables and join keys.** `product_reviews` joined to `deliveries` on `order_id` (many-to-one); the review count remains 7,000. **Why this visualisation.** Rating means with 95% intervals, full rating composition and operational subgroups test whether a small mean difference hides a distributional change. **Interpretation and explanation.** OTIF, delay days, service level and delay reason do not meaningfully separate either mean rating or the full 1–5 star mix in the reviewed orders. Delivery performance is therefore a weak rating discriminator in this dataset. **Material limitation.** Some orders contribute more than one review, and orders without reviews are not represented.

3. Evidence-based findings

3.1 **The median gives the clearest view of a typical order.** Across 5,000 orders, Figure 1 shows a median total of **$2,555**, a mean of **$3,000** and a 99th percentile near **$9,832**. The high-value tail lifts the mean, so routine basket reporting should present the median together with an upper percentile. Part of the spread reflects differences in the number of items per order.

3.2 **Order volume is balanced across the three sales channels.** Figure 1 shows Web, Store and Mobile each contributing approximately **32–35% of 5,000 orders**. The business is therefore not dependent on one channel for transaction volume. Revenue, margin and acquisition cost are not included in this comparison and should be added before making channel-investment decisions.

3.3 **Customer segment and loyalty tier do not meaningfully distinguish basket value.** Across 5,000 orders linked to 500 customers, Figure 2 shows segment medians within **$203** and loyalty-tier means between **$2,967 and $3,038**, with overlapping 95% intervals. Basket-value targeting should therefore place more weight on order context and product mix. Because customers place repeated orders, a customer-clustered comparison would refine the interval estimates.

3.4 **Order frequency is the clearest separator of customer revenue in 2018.** At customer grain (500 customers), Figure 2 shows cumulative revenue rising closely with recorded order count, while loyalty tiers show little separation. This makes future purchase frequency a practical retention target. The relationship reflects one period of accumulated revenue, so a later time window is needed to test whether historical frequency predicts future activity.

3.5 **Price point, rather than unit volume alone, drives the category revenue ranking.** Across 15,723 order-item lines and 41,993 units, Figure 3 shows Home Entertainment and Laptop producing **34.4% of $16.12m revenue from 14.4% of units**. Accessory records the largest unit share but only about **$0.61m** revenue. Assortment planning should compare units, revenue and margin together; this figure does not include returns or realised profit.

3.6 **Order totals must be calculated before expanding orders to item grain.** The grain-control statistic computed alongside Figure 3 shows that summing `order_total` after the orders–items join produces **$53.19m**, which is **3.55 times** the correct **$15.00m** total. Pre-aggregation and reconciliation to the source total are therefore required for every order-level revenue calculation. This is a data-grain finding rather than a customer-demand result.

3.7 **The main demand pattern is intraday rather than monthly or weekly.** Across 5,000 orders in 2018, monthly counts range from **364 in February to 438 in January**, while weekday and weekend rates remain similar. February also has fewer calendar days, so its lower count alone is not evidence of seasonality. Orders concentrate in trading hours and fall sharply overnight, making hour of day the clearest staffing signal in this dataset; another year and a confirmed timezone would show whether the pattern persists.

3.8 **Writing style explains review length more clearly than star rating.** Among 7,000 reviews, Figure 5 shows **64.9%** at 4–5 stars and **17.6%** at 1–2 stars. Median length rises from about **433 characters** for concise reviews to **1,572** for narrative reviews, while length and helpful votes remain broadly flat across ratings. Text analysis should therefore control for writing style; helpful-vote comparisons also need a views or exposure denominator.

3.9 **Whitespace word counts are not comparable across writing systems.** Figure 6 covers 7,000 reviews, including **303 non-Latin-script reviews**, and shows a different character-to-token relationship by script class. Cross-language text features should use character measures or language-aware tokenisation, with results reported by language or script group. The non-Latin group combines several writing systems, so language-specific estimates require larger samples.

3.10 **Delivery performance is a weak discriminator of star rating in this dataset.** Figure 7 reports **89.3% OTIF across 5,000 deliveries** with a **3.7 percentage-point** carrier spread, while Figure 8 shows mean ratings of about **3.73 for late** and **3.71 for on-time** reviews, with overlapping intervals across 7,000 reviews. Delivery teams should manage OTIF as an operational outcome rather than relying on star rating as its proxy. The rating comparison covers reviewed orders and includes multiple reviews for some orders.

4. Future machine-learning questions

No model is trained in this assignment. Each question is grounded in an EDA result and uses predictors available at the proposed decision time.

4.1 OTIF-failure classification

Use case, unit and target. Before dispatch, rank **orders** for intervention; target `on_time_in_full = False`.

Decision-time predictors. Warehouse, carrier, service level, sales channel, expedited flag, order timestamp, item count/value, product weight and destination fields available before dispatch.

Validation and metrics. Forward time split by order date; PR-AUC and recall at a fixed intervention capacity, with ROC-AUC secondary.

Risk and EDA grounding. Exclude delivered date, delay days/reason and realised fulfilment fields. Carrier assignment may encode operational policy. Figure 7 grounds the 10.7% event rate and weak simple group separation.

4.2 Fulfilment-hours regression

Use case, unit and target. At order acceptance, estimate **hours per order** to support workload promises; target `fulfilment_hours`.

Decision-time predictors. Order time, warehouse, channel, service level, expedited flag, item count, basket value, product mix, total weight and destination features available at acceptance.

Validation and metrics. Rolling/forward time split; MAE plus 90th-percentile absolute error.

Risk and EDA grounding. Do not use dispatch/delivery dates or post-dispatch delay fields. Process changes can cause temporal drift. Figure 7 shows that distance bands and expedited status alone explain little variation.

4.3 Low-rating classification

Use case, unit and target. At review invitation time, identify **reviewed orders** at risk of a 1–2 star rating; target whether order-level mean rating is at most 2.

Decision-time predictors. Pre-invitation order, product/category, customer-history and completed-delivery fields; text and the eventual rating are excluded.

Validation and metrics. Forward split by delivery/review date, grouped by customer; PR-AUC, Brier score and recall at fixed contact capacity.

Risk and EDA grounding. Review propensity creates selection bias; sensitive customer proxies require subgroup checks. Figures 5 and 8 show class imbalance and weak delivery-only separation.

4.4 Future customer-frequency regression

Use case, unit and target. At a period boundary, forecast **next-period order count per customer** for retention planning; target future count.

Decision-time predictors. Only customer attributes and aggregates from periods ending before the boundary: recency, historical frequency/value, channel mix, category breadth and prior delivery experience.

Validation and metrics. Rolling-origin temporal validation; MAE and Poisson deviance, compared with a recency-frequency baseline.

Risk and EDA grounding. Random splits and lifetime fields can leak the future; new customers need a fallback. Figure 2 shows current revenue is closely associated with frequency but tier alone adds little visible separation.

4.5 Product clustering

Use case, unit and target. Group **products** for assortment and inventory review; unsupervised objective: stable, interpretable product segments.

Decision-time predictors. Catalogue attributes plus pre-cut-off price, cost, weight and historical unit/revenue summaries; scale numeric features and encode category/brand carefully.

Validation and metrics. Fit on an earlier period and assess stability on a later period using adjusted Rand index; silhouette score and business interpretability as secondary criteria.

Risk and EDA grounding. Dominant category labels and scaling can predetermine clusters; rare products may be unstable. Figure 3 motivates separating high-price/low-volume from low-price/high-volume products.

5. Limitations and conclusion

The eight figures cover all six required EDA categories, use all six tables and include four checked relational analyses. The strongest practical results are the importance of table grain, the separation between category price and volume, the concentration of demand by hour of day, and the limited value of broad customer labels for basket targeting. The analysis covers one 2018 export; repeated customers, multiple reviews per order and review-selection effects are addressed in the relevant figure limitations.

References

National Institute of Standards and Technology. (n.d.). *Histogram*. In *NIST/SEMATECH e-Handbook of statistical methods*. https://www.itl.nist.gov/div898/handbook/eda/section3/histogra.htm

External material informed general chart selection only. All numerical claims, findings and ML motivations above are calculated from the six Group030 standardised tables.